

DS351L - Data Visualization Lab

Lab 09

23rd April 2026

Create a Tableau Story using:

- at least 3 worksheets
- at least 1 dashboard
- a dataset of your own choice

Your story should present a meaningful analysis, not random charts.

Assume you are working as a Data Visualization Analyst. Your manager has asked you to prepare a visual presentation for stakeholders who do not have time to read raw tables or lengthy reports. They want a story that:

- starts with an overview
- highlights important comparisons or trends
- explains one or more key findings
- ends with a dashboard that allows interactive exploration

You must think like an analyst, not just a software user.

Dataset

You may choose any dataset of your choice, but it must satisfy these conditions:

- it must contain enough data for analysis
- it must allow creation of at least 3 meaningful charts
- it must support filtering and comparison
- it should contain fields that can help answer a real question

Examples of suitable datasets:

- sales and profit dataset
- e-commerce data
- student performance data
- healthcare data
- sports data
- airline or transport data
- environmental data
- population data
- employment data
- movie or streaming data

Do not choose a dataset that is too small or too simple.

You are required to create the following in Tableau:

1. Three Worksheets

Create at least 3 different sheets.

Each sheet must:

- answer a meaningful question
- use an appropriate chart type
- have a clear and professional title
- be visually appealing
- use colors meaningfully, not randomly
- be easy to read and understand

Possible examples:

- trend over time
- comparison across categories
- geographical pattern
- ranking of top or bottom performers
- relationship between two variables

The charts must not be repetitive. Each chart should contribute something different to the overall analysis.

2. One Dashboard

Create at least 1 dashboard using your worksheets.

The dashboard must:

- include multiple sheets in a clean layout
- include filters
- have a meaningful dashboard title
- be visually balanced and readable
- help the user interact with the data

The dashboard should look like something a decision-maker can use to explore the findings further.

3. One Tableau Story

Create a Tableau Story based on your sheets and dashboard.

Your story must:

- include multiple story points

- include captions
- use colors properly
- guide the audience from one insight to the next
- end with the dashboard as an interactive summary

The story should feel like a guided presentation.

Detailed Expectations!!

A. Meaningful Analysis

Your work should answer real questions such as:

- What is happening?
- Where is the problem?
- Which category performs best or worst?
- Is performance changing over time?
- What should management pay attention to?

Do not create charts just because Tableau allows them.

B. Chart Selection

Choose chart types wisely.

Examples:

- line chart for trends over time
- bar chart for category comparison
- map for geographical data
- scatter plot for relationships
- stacked or side-by-side bars for subgroup comparison

Avoid poor chart choices that make interpretation difficult.

C. Titles

All worksheets, dashboard, and story points must have good titles.

Bad title:

- Sheet 1
- Sales Chart
- Dashboard

Good title:

- Monthly Sales Trend from 2022 to 2024
- Profit Comparison Across Product Categories
- Regional Performance Dashboard
- Story Point: West Region Shows Highest Growth

Titles should clearly tell the viewer what they are looking at.

D. Captions

Each story point must include a caption.

A caption should explain the key takeaway, not just repeat the chart title.

Bad caption:

- This chart shows sales by category.

Good caption:

- Technology contributes the highest profit, while Furniture shows weaker profitability despite strong sales.

E. Use of Color

Colors must be used with purpose.

Use color to:

- distinguish categories
- highlight top and bottom performers
- emphasize important insights
- make charts readable and attractive

Do not use too many random colors.

Keep the dashboard consistent.

F. Visual Appeal

Your work should look polished and professional.

This includes:

- neat alignment
- readable font sizes
- proper spacing

- no clutter
- balanced layout
- attractive but not excessive formatting

Think of it as something you would present to a client or management team.